

# Hierarchical Graph Neural Network for Interpretable Facial Weakness Assessment in Facioscapulohumeral Muscular Dystrophy

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## Abstract

**Background:** Facioscapulohumeral muscular dystrophy (FSHD) is among the most prevalent muscular dystrophies, characterized by initial asymmetric weakness of facial and shoulder girdle muscles. Early detection remains challenging due to the subtle onset of facial weakness, which currently relies on subjective and variable clinical assessment. The lack of objective and scalable measures hampers timely intervention. This study investigated an automated FSHD facial assessment, which includes a 3-level grading of facial weakness and the prediction of facial weakness asymmetry using static facial photographs.

**Methods:** We collected eight expression photographs from 46 FSHD patients and 52 healthy controls. To protect patient privacy, de-identified facial landmarks were detected and encoded into action units and asymmetry keys. We developed a hierarchical graph neural network (HGNN) to integrate clinical priors across facial regions, actions, and asymmetry into three hierarchical graphs for multi-task classification.

**Results:** Model interpretation based on Shapley values revealed that HGNN prioritized facial features consistent with expert clinicians. The HGNN-derived facial weakness scores achieved task-level consistency with expert clinicians of 76.1% and 74.9% in the two tasks. The class-specific accuracies are 87.8%, 80.8%, 83.5% for absence, moderate, and severe facial weakness grading, and 88.1%, 80.1%, 81.6% for non-asymmetry, left-, and right-side predominant facial weakness conditions, respectively.

**Conclusions:** These findings provide proof of principle that graph-based clinical knowledge can support automatic assessment of facial weakness in FSHD using standardized facial photographs. Our study confirms the potential of 2D data to provide reliable facial evaluation for routine follow-up studies.

## Plain Language Summary

Facioscapulohumeral muscular dystrophy (FSHD) is one of the most prevalent muscular dystrophies, characterized by initial asymmetric weakness of facial muscles. Early detection remains challenging due to the

subtle onset of facial weakness, which currently relies on self-reporting and physician-reported clinical assessment that can vary between observers. Our study shows it is possible to automatically detect and assess facial weakness in patients with FSHD using images of their faces showing eight different expressions. We demonstrate that a machine learning approach—specifically, a graph-based model that incorporates clinical knowledge—can effectively analyze facial features relevant to FSHD. It highlights that leveraging standardized facial photos and advanced computational models can provide a reliable, objective tool for assessing facial weakness in FSHD.

## Introduction

Facioscapulohumeral Muscular Dystrophy (FSHD) is among the most common inherited muscular dystrophies and is typically transmitted in an autosomal dominant pattern [1, 2]. Reported prevalence estimates range from 1 per 8,300 to 1 per 1,300,000 individuals [3, 4, 5, 6]. FSHD usually presents with progressive, often asymmetrical weakness that begins in the facial and shoulder girdle muscles and later involves other muscle groups, including the lower limbs and, in some cases, respiratory muscles, potentially leading to wheelchair dependence or a need for ventilatory support [7].

Genetic analysis constitutes the gold standard for confirming FSHD [8], whereas clinical and imaging assessments characterize its phenotypic manifestations. Magnetic resonance imaging (MRI) [9, 10], ultrasonography [11], and dual-energy X-ray absorptiometry (DEXA) [12], together with machine learning algorithms [13, 14], support quantitative grading [11] and progression monitoring [15]. However, equipment requirements and operational complexity limit their accessibility for population-level screening. A clinical need persists for automated, cost-effective, and objective diagnostic solutions for facial weakness assessment in FSHD.

Facial weakness is one of the first and most characteristic features of FSHD [16, 17]. Using facial features to assess FSHD may facilitate earlier detection of the presentation compared to methods based on the lower limbs [9, 10, 11, 13, 14, 15]. Yet, facial weakness is usually subtle and easily overlooked. Individuals may have difficulty fully closing the eyes; asymmetrical smiling; reduced expressiveness of facial movements; diminished facial muscle tone with a flattened appearance [7, 16]. Weakness of the orofacial muscles, critical for articulation and airflow control, leads to difficulty producing certain speech sounds and reduced speech clarity. Facial muscle weakness can also impair eating and drinking, and limit the use of facial expressions for non-verbal communication, causing psychological stress [18, 19].

Recognizing FSHD-related facial weakness is challenging for both affected individuals and clinicians. Subtle changes may first be noticed by family members, friends, or healthcare professionals. Manual clinical assessment remains time-intensive and subjective. Clinical severity is commonly described using the Ricci score [20] or the FSHD clinical score [21]. The Comprehensive Clinical Evaluation Form further supports standardized phenotypic classification [22]. Loonen *et al.* developed a four-point facial weakness score, but reported limited interobserver agreement [23]. Similar weaknesses in interobserver reliability have been observed in other facial assessments [24, 25]. Differences between patient-reported facial function and clinician-rated weakness indicate that these measures capture complementary aspects of disease burden [26, 27]. These limitations motivate more objective and reproducible assessment of FSHD facial involvement [28, 29, 30].

Recent work has explored imaging and computational approaches to this need [31]. Hangul *et al.* used three-dimensional morphometric scanning to quantify facial movements, identifying differences between patients and controls in oral expressions and eye closure [29]. Loonen *et al.* developed a convolutional neural network (CNN) that estimates facial weakness from neutral and maximum-movement frames from facial exercise videos [32]. Berges *et al.* related objective facial movement measurements, including dental display and commissure excursion, to patient-reported FACE-Q outcomes [28]. Vincenten *et al.* [30] applied ultrasound as an alternative to manual muscle testing for facial muscle structural quantification in FSHD patients. These studies establish the clinical value of characterizing facial involvement and motivate accessible assessments that relate observable facial function to clinically meaningful severity.

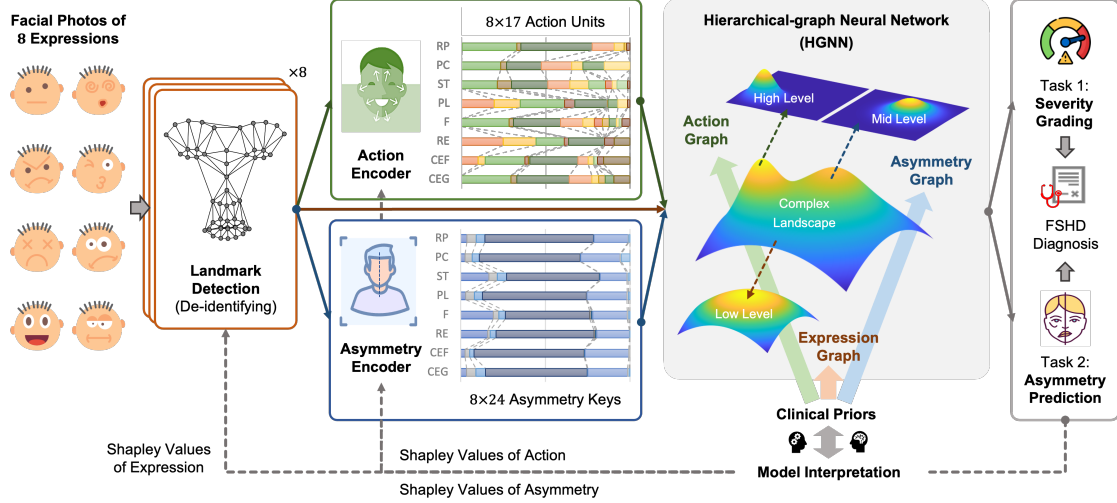


Figure 1: Overview of the proposed method for facial weakness assessment in FSHD. The pipeline involves photo collection, landmark detection, feature encoding, a hierarchical graph neural network (HGNN) for multi-task prediction, and Shapley value-based model interpretation.

Facial muscle involvement caused by FSHD bears unique facial characteristics that differ from those seen in common unilateral facial nerve palsy, stroke, or bilateral facial weakness in other myopathies or dystrophies. In FSHD, weakness typically affects both sides of the face, but is often asymmetrical. The most commonly affected facial muscles in FSHD include depressor anguli oris, orbicularis oris, orbicularis oculi, zygomaticus, buccinator, and even frontalis and corrugator [30, 33]. On the other hand, the levator palpebrae superioris, the extraocular eye muscles, and the mastication muscles, such as masseter and temporalis, are never affected. The degree of muscle involvement varies among patients and can affect different regions of the face to varying extents [23]. Facial weakness can also be minimal or absent in up to 25% of patients [16]. Thus, an accurate assessment typically requires professional, experienced physicians. Because FSHD is rare, patient recruitment and high-quality data collection are time-consuming. Although machine learning methods perform well on general facial recognition tasks [34], developing robust models for FSHD facial assessment under data scarcity remains a challenge [35, 36].

Graph-based learning [37] offers a way to model non-Euclidean topology in facial geometry, with clinical applications in pain estimation [38] and depression detection [39]. Encoded facial landmarks and action units (AUs) reduce reliance on identifiable images to protect privacy, while their anatomical and functional meanings support clinical prior integration and model interpretation. Facial region-based approaches include Region Attention Networks for facial expression recognition [40] and learned connectivity patterns between facial regions [41]. Action graphs provide complementary motion information to facial landmarks. Jin *et al.* [42] modeled dependencies among 20 facial regions linked to specific AUs; Liu *et al.* [43] formulated AU detection as graph relation modeling; and Lei *et al.* combined facial graphs with AU features for micro-expression recognition [44]. However, adapting these approaches to FSHD requires explicit consideration of disease-specific expression patterns, bilateral asymmetry, and clinical grading criteria.

Building on prior efforts to characterize facial phenotypes in FSHD [23, 27, 45] and graph-based facial analysis, we developed an interpretable algorithm to assess FSHD-related facial weakness, as shown in Fig. 1. We constructed three hierarchical graphs defined by neurologists to embed clinical priors. The graph representation regulates the model to learn in a way consistent with clinical patterns validated by three experienced neurologists specializing in neuromuscular diseases. Our method demonstrates the feasibility of privacy-preserving, automated facial feature extraction in FSHD, enabling both the grading of facial weakness severity and the prediction of facial weakness asymmetry from only facial photographs. The

clinician-involved model development and clinician-validated model interpretation help foster trust in the automated FSHD assessment.

## Methods

As Fig. 1, our workflow involves four stages. (1) **Subject recruitment**: We recruited FSHD patients and healthy controls. All participants were photographed with eight facial expressions [23]. (2) **Facial feature encoding**: To ensure patient privacy, raw photos were processed into de-identified action units and facial asymmetry keys. (3) **Model Architecture**: A hierarchical-graph neural network (HGNN) is developed to predict the severity and asymmetry of FSHD. (4) **Results**: Model performance was evaluated through cross-validation and Shapley value interpretation [27].

### Subject recruitment

Recruitment lasted five years between June 2021 and May 2026. FSHD patients were identified via the Hong Kong Neuromuscular Disorder Patient Registry, covering all public hospitals in Hong Kong [46]. Diagnoses were confirmed by clinical neurologists and genetic testing [47, 48].

Photographs were captured from the front face of each subject [23]. The image acquisition protocol is described in S1 of the supplementary material. Participants were instructed to perform eight standardized expressions as an expression set ( $|\phi_e| = 8$ ): resting position (E-RP), closing the eyes gently (E-CEG), closing the eyes firmly (E-CEF), raising the eyebrows (E-RE), frowning (E-F), pursing the lips (E-PL), showing the teeth (E-ST), and puffing the cheeks (E-PC). For each genetically confirmed FSHD patient, facial weakness severity and asymmetry labels were assigned through consensus review by any two of the three experienced neurologists. This labeling criterion followed the validated FSHD Evaluation Scale (Comprehensive Clinical Evaluation Form, Section 2) [21, 22], which distinguishes the facial weakness between partial ability and inability to perform facial tasks.

### Facial feature encoding

**Action Encoder**: Photographs were processed using OpenFace 2.0 [49], PyFeat [50], and Face Alignment Network (FAN) [51] to detect de-identified facial landmarks. Trained clinicians visually inspected three landmark overlays and selected the most anatomically plausible one without motion blur. The detailed landmark detection performance is described in S2 of the supplementary material. Action units (AUs) were extracted using the valid landmarks based on the Facial Action Coding System [52]. Following z-score normalization and frontal alignment, 17 AUs were encoded ( $|\phi_{au}| = 17$ ): inner/outer brow raiser (AU-IBR/OBR), brow lowerer (AU-BL), upper lid raiser (AU-ULidR), cheek raiser (AU-CheR), lid tightener (AU-LidT), nose wrinkler (AU-NW), upper lip raiser (AU-ULipR), lip corner puller/depressor (AU-LCP/LCD), dimpler (AU-D), lip stretcher/tightener (AU-LS/LipT), lips part (AU-LP), chin raiser (AU-ChiR), jaw drop (AU-JD), and blink (AU-B).

Table 1: The labeling criteria of facial weakness severity (CCEF score) and asymmetry for FSHD patients

| Score | Category      | Evaluation Criteria                                                                                        |
|-------|---------------|------------------------------------------------------------------------------------------------------------|
| 0     | Absence       | Absence of facial weakness                                                                                 |
| 1     | Moderate      | Partial ability to complete at least one of the expressions: close eyes, protrude lips, or puff out cheeks |
| 2     | Severe        | Unable to complete at least one of the expressions: close eyes, protrude lips, or puff out cheeks          |
| /     | Non-asymmetry | Equal weakness on both sides of the face                                                                   |
| /     | Left          | Left-side face presents higher weakness                                                                    |
| /     | Right         | Right-side face presents higher weakness                                                                   |

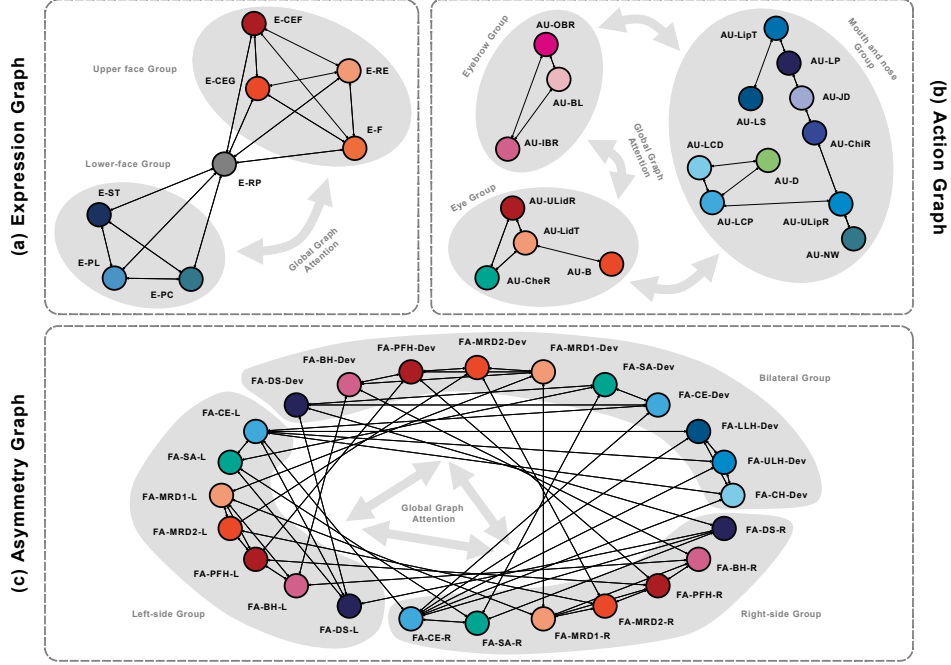


Figure 2: The definition of three graphs. (a) Expression graph, where “E-xxx” nodes stand for 8 expressions. (b) Action graph, where “AU-xxx” nodes represent 17 action units. (c) Asymmetry graph, where “FA-xxx-L”, “FA-xxx-R”, and “FA-xxx-Dev” nodes represent 7 left-side, 7 right-side, and 10 deviation asymmetry keys.

**Asymmetry Encoder:** Geometric measurements derived from facial landmarks yielded 24 asymmetry keys ( $|\phi_{fa}| = 24$ ): seven unilateral keys per side and ten bilateral keys. Specifically, the commissure excursion (facial asymmetry-commissure excursion-left/right abbr. FA-CE-L/R) is the distance from mouth to lip corners; Smile angle (FA-SA-L/R) is the vertical position of commissures relative to the mouth center; Marginal reflex distances are from the pupil center to the upper (FA-MRD1-L/R) and lower (FA-MRD2-L/R) eyelids; Brow height (FA-BH-L/R) is the vertical distance from pupil to eyebrow; Palpebral fissure height (FA-PFH-L/R) is the vertical distance between the upper and lower eyelids; Dental show (FA-DS-L/R) is the distance between the upper and lower lips at the midline. Deviations between these unilateral keys are also calculated and named FA-CE-Dev, FA-SA-Dev, FA-MRD1/2-Dev, FA-BH-Dev, FA-PFH-Dev, and FA-DS-Dev. Besides, commissure height deviation (FA-CH-Dev) measures the vertical distance between the left and right commissures; Upper/lower lip height deviation (FA-ULH/LLH-Dev) measures the height difference of the upper/lower lips between two sides.

## Hierarchical-graph neural network

The collected data is analyzed from three perspectives using hierarchical graphs: low-level expression, mid-level landmark asymmetry, and high-level action. As shown in Fig. 1, each perspective provides unique insights into facial weakness and contributes to a comprehensive spectrum of affected facial muscles in FSHD.

### Graph representation

**Expression Graph:** The low-level expression graph  $\mathcal{G}_e$  focuses on the eight facial expressions, capturing holistic facial movement and function. Node set  $\phi_e$  is categorized by upper and lower face regions: In the upper face, E-CEG and E-CEF engage the orbicularis oculi, while transitions between E-CEF and E-F, as well as E-CEG and E-RE, capture synergistic eye-forehead muscle interactions; the antagonistic relationship

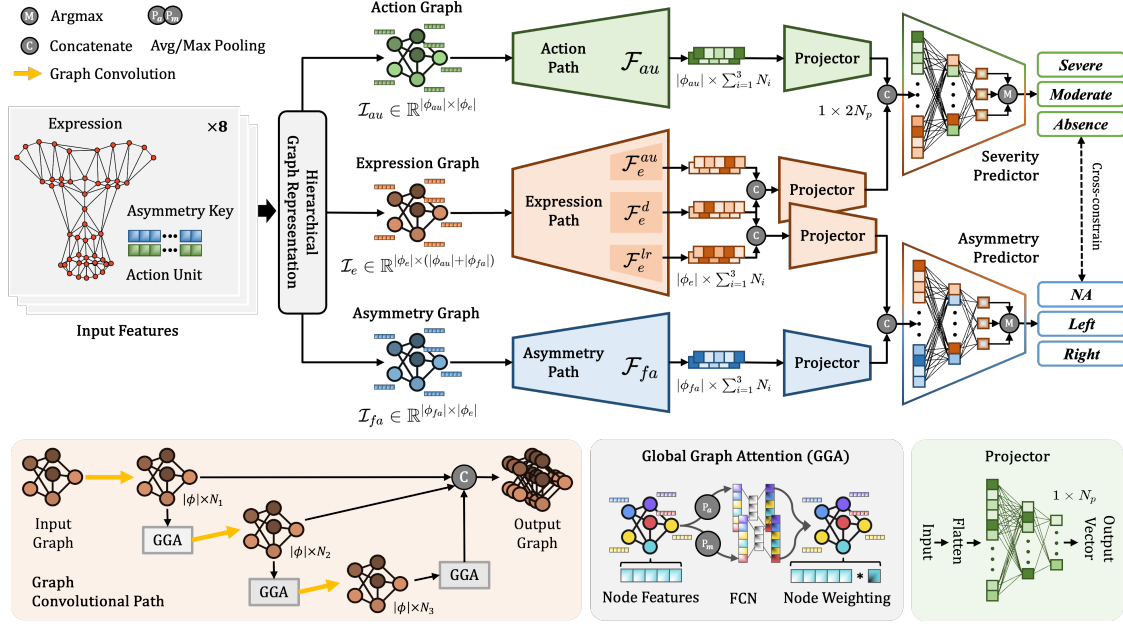


Figure 3: Hierarchical-graph neural network. The inputs are action units and asymmetry keys of 8 expressions, represented as the action graph  $\mathcal{I}_{au}$ , asymmetry graph  $\mathcal{I}_{fa}$ , and expression graph  $\mathcal{I}_e$  for the action path ( $\mathcal{F}_{au}$ ), asymmetry path ( $\mathcal{F}_{fa}$ ), and expression path (including action  $\mathcal{F}_e^{au}$ , deviation  $\mathcal{F}_e^d$ , left- and right-side  $\mathcal{F}_e^{lr}$  branches).  $|\phi_e| = 8$ ,  $|\phi_{au}| = 17$ , and  $|\phi_{fa}| = 24$  denote the numbers of expression, action, and asymmetry nodes.  $N_i = 32, 64, 128$  and  $N_p = 128$  denote output feature number of the  $i$ -th graph convolutional layer and the projector.

between E-RE and E-F reflects frontalis and corrugator dynamics. In the lower face, E-ST, E-PC, and E-PL involve the zygomaticus complex, representing coordinated perioral muscle activation. Guided by these physiological relationships, coupled expressions are connected via the edge set  $\gamma_e$ . E-RP serves as a neutral baseline and is linked to all active expressions. The resulting expression graph is defined as  $\mathcal{G}_e = (\phi_e, \gamma_e)$  in Fig. 2(a).

**Asymmetry Graph:** The mid-level asymmetry graph  $\mathcal{G}_{fa}$  focuses on the bilateral facial landmarks and provides a structured representation of their interrelationships. As Fig. 2 (c), the edge set  $\gamma_{fa}$  links deviation keys to their corresponding unilateral keys (e.g., FA-CE-Dev  $\leftrightarrow$  FA-CE-L/R) and paired left-right counterparts (e.g., FA-CE-L  $\leftrightarrow$  FA-CE-R).  $\mathcal{G}_{fa} = (\phi_{fa}, \gamma_{fa})$  underscores the complex interplay between contralateral facial features, enabling the model to quantify facial symmetry.

**Action Graph:** The high-level action graph  $\mathcal{G}_{au}$  focuses on the node set  $\phi_{au}$ , modeling muscle groups that are coordinated to execute a given expression. This graph encodes anatomical contingencies across three functional clusters: (1) The eyebrow group, including AU-IBR, AU-OBR, and AU-BL, characterizes the reciprocal biomechanical coupling between frontalis elevation and depressor supercilli activity. (2) The eye group, including AU-ULidR, AU-CheR, AU-LidT, AU-B, delineates the kinematic synergy within the periocular complex, linking palpebral fissure modulation with infraorbital muscle contraction. (3) The mouth and nose group, including AU-NW, AU-ULipR, AU-LCP, AU-D, AU-LCD, AU-ChiR, AU-JD, integrates the oral-nasal complex. This group reveals the physiological dependencies between the perioral muscles regulating lip corner movement and dimple formation, and also between mandibular depression and contraction of the mandibular muscles. These actions collectively constrain the vertical dimension of the face and determine lip posture. The edge set  $\gamma_{au}$  establishes topology within these tripartite subsets, defining  $\mathcal{G}_{au} = (\phi_{au}, \gamma_{au})$  as Fig. 2 (b).

## Model Architecture

Figure 3 shows the proposed hierarchical-graph neural network (HGNN). The first stage transforms expression, action, and asymmetry features into three embeddings via graph convolutional layers. The second stage aggregates the severity-related vectors to assess the FSHD score via a severity predictor; meanwhile, it aggregates the asymmetry-related vectors to predict facial weakness asymmetry via an asymmetry predictor.

**Hierarchical Graph Perception:** For each subject, the model input is  $\mathcal{I}_e \in \mathbb{R}^{|\phi_e| \times (|\phi_{au}| + |\phi_{fa}|)}$ .  $\phi_{au}$  and  $\phi_{fa}$  constitute sub-nodes of  $\phi_e$ ,  $\mathcal{G}_{au}$  and  $\mathcal{G}_{fa}$  are sub-graphs of  $\mathcal{G}_e$ . HGNN builds three paths for three hierarchical graphs: expression path ( $\mathcal{F}_e$ ), action path ( $\mathcal{F}_{au}$ ), and asymmetry path ( $\mathcal{F}_{fa}$ ). HGNN first reorganizes  $\mathcal{I}_e$  into  $\mathcal{I}_{au} \in \mathbb{R}^{|\phi_{au}| \times |\phi_e|}$  as the input of  $\mathcal{F}_{au}$  and  $\mathcal{I}_{fa} \in \mathbb{R}^{|\phi_{fa}| \times |\phi_e|}$  as the input of  $\mathcal{F}_{fa}$ , where  $\mathcal{I}_e = \mathcal{I}_{au} \cup \mathcal{I}_{fa}$  remains the input of  $\mathcal{F}_e$ . In other words, the values of an action unit, derived from all expressions, are initialized as features of that action node; the values of an asymmetry key, derived from all expressions, are initialized as features of that asymmetry node.

In  $\mathcal{F}_e$ ,  $\mathcal{G}_e$  only constrains the high-level  $\phi_e$ , where sub-nodes  $\phi_{au}$  and  $\phi_{fa}$  are regarded as separable features. Because AU computing is isotropic and implies no asymmetry information,  $\phi_{au}$  features ( $\mathcal{I}_e^{au} \in \mathbb{R}^{|\phi_e| \times |\phi_{au}|}$ ) should not contribute to asymmetry prediction. Severity scoring involves landmark deviation between two facial sides, rather than unilateral behaviors. So  $\phi_{fa}$  features are spited into two parts: a deviation part  $\mathcal{I}_e^d \in \mathbb{R}^{|\phi_e| \times |\phi_d|}$  for severity prediction, and a unilateral part  $\mathcal{I}_e^{lr} \in \mathbb{R}^{|\phi_e| \times |\phi_{lr}|}$  for asymmetry prediction. Accordingly, HGNN divides  $\mathcal{F}_e$  into three branches:  $\mathcal{F}_e^{au}$  with input  $\mathcal{I}_e^{au}$ ,  $\mathcal{F}_e^d$  with input  $\mathcal{I}_e^d$ , and  $\mathcal{F}_e^{lr}$  with input  $\mathcal{I}_e^{lr}$ . Herein, HGNN embeds rich priors into the model architecture, which is crucial for rare diseases under limited data availability.

**Embedding Path:** As shown in Fig. 3,  $\mathcal{F}_{au}$ ,  $\mathcal{F}_{fa}$ ,  $\mathcal{F}_e^{au}$ ,  $\mathcal{F}_e^{lr}$ , and  $\mathcal{F}_e^d$  share an identical 3-layer architecture. For clarity, let  $\mathcal{G}$  denote the active graph for node set  $\phi$ . At layer  $i$ , HGNN processes node features  $\phi$  ( $i = 1$ ) or their updated vectors ( $i = 2, 3$ ) alongside  $\mathcal{G}$ . The feature vector  $\mathbf{v}_n^i$  for node  $n \in \phi$  is computed by aggregating its first-order neighbors  $\mathcal{N}(n|\mathcal{G})$  and its prior state [53], which collectively define the node’s receptive field. A graph convolution is formulated as

$$\begin{aligned} \mathbf{v}_{\mathcal{N}(n|\mathcal{G})}^i &= \frac{1}{|\mathcal{N}(n|\mathcal{G})|} \sum_{\tilde{n} \in \mathcal{N}(n|\mathcal{G})} \mathbf{v}_{\tilde{n}}^{i-1}, \\ \mathbf{v}_n^i &= \alpha \left( \mathbf{W}_{\mathcal{G}}^i * \left[ \mathbf{v}_{\mathcal{N}(n|\mathcal{G})}^i; \mathbf{v}_n^{i-1} \right] \right), \end{aligned} \quad (1)$$

where  $\mathbf{v}_{\mathcal{N}(n|\mathcal{G})}^i$  denotes the aggregated information,  $\mathbf{W}_{\mathcal{G}}^i$  denotes the learnable weights,  $\alpha(\cdot)$  denotes layer normalization and GeLU activation,  $[\cdot; \cdot]$  denotes concatenation. HGNN employs graph convolution on  $\forall n \in \phi$  to upscale feature dimension to 32, 64, 128 in three layers.

HGNN adopts mean aggregation with homogeneous neighbor weights, yet clinical criteria indicate heterogeneous node importance. For instance, E-CEF, E-ST, and E-PC, alongside asymmetry in the lower face, are more indicative of FSHD symptoms. We propose a global graph attention (GGA) block to address this node imbalance by global weighting across the graph after each layer. It first performs global pooling on features in  $\forall n \in \mathcal{G}$  and stacks all nodes into one-dimensional  $\mathbf{v}_{max}^i, \mathbf{v}_{avg}^i \in \mathbb{R}^{1 \times |\phi|}$ :

$$\begin{aligned} \mathbf{v}_{max}^i &= [P_{max}(\mathbf{v}_n^i) \mid \forall n \in \phi]^T, \\ \mathbf{v}_{avg}^i &= [P_{avg}(\mathbf{v}_n^i) \mid \forall n \in \phi]^T, \end{aligned} \quad (2)$$

where  $P_{avg}(\cdot)$  and  $P_{max}(\cdot)$  denote average and max pooling. A fully connected layer computes global attention weights  $\mathcal{A}^i$  for layer  $i$  to modulate the graph convolution output:

$$\begin{aligned} \mathcal{A}^i &= \sigma \left( \mathbf{W}_2^i \cdot \alpha \left( \mathbf{W}_1^i \cdot \mathbf{v}_{avg}^i \right) + \mathbf{W}_2^i \cdot \alpha \left( \mathbf{W}_1^i \cdot \mathbf{v}_{max}^i \right) \right), \\ [\mathbf{v}_n^i \mid \forall n \in \phi] &= \mathcal{A}^i \odot [\mathbf{v}_n^i \mid \forall n \in \phi]^T, \end{aligned} \quad (3)$$

where  $\sigma(\cdot)$  is the sigmoid function,  $\mathbf{W}_1^i$  and  $\mathbf{W}_2^i$  are weights of two fully connected layers with  $\frac{|\phi|}{4}$  and  $|\phi|$  channel numbers, respectively.

HGNN extracts facial features from three graphs using hierarchical perception and generates five embeddings:

$$\begin{aligned}\mathcal{V}_{au} &= \mathcal{F}_{au}(\mathcal{I}_{au}|\mathcal{G}_{au}) \in \mathbb{R}^{|\phi_{au}| \times \sum_{i=1}^3 N_i}, \\ \mathcal{V}_{fa} &= \mathcal{F}_{fa}(\mathcal{I}_{fa}|\mathcal{G}_{fa}) \in \mathbb{R}^{|\phi_{fa}| \times \sum_{i=1}^3 N_i}, \\ \mathcal{V}_e^{au/lr/d} &= \mathcal{F}_e^{au/lr/d}(\mathcal{I}_e^{au/lr/d}|\mathcal{G}_e) \in \mathbb{R}^{|\phi_e| \times \sum_{i=1}^3 N_i},\end{aligned}\tag{4}$$

where each vector concatenates three-layer outputs along the node feature axis.

**Predictor:** A projector  $\mathcal{F}_p(\cdot)$  maps these vectors into a uniform feature space ( $N_p = 128$ ). Based on clinical priors,  $\mathcal{V}_e^{au}$  lacks asymmetry information, while unilateral  $\mathcal{V}_e^{lr}$  lacks severity information. Thus,  $\mathcal{V}_e^{au}$  and  $\mathcal{V}_e^d$  are projected together to predict severity;  $\mathcal{V}_e^{lr}$  and  $\mathcal{V}_e^d$  are projected together to predict asymmetry.

Predictors  $\mathcal{F}_s(\cdot)$  and  $\mathcal{F}_a(\cdot)$  co-learn severity and asymmetry tasks, sharing the  $\mathcal{F}_p(\cdot)$  architecture. Predictions are formulated as

$$\begin{aligned}[\mathcal{P}_s^s, \mathcal{P}_s^m, \mathcal{P}_s^a]^T &= \rho(\mathcal{F}_s([\mathcal{F}_p(\mathcal{V}_{au}); \mathcal{F}_p([\mathcal{V}_e^{au}; \mathcal{V}_e^d])])), \\ [\mathcal{P}_a^n, \mathcal{P}_a^l, \mathcal{P}_a^r]^T &= \rho(\mathcal{F}_a([\mathcal{F}_p(\mathcal{V}_{fa}); \mathcal{F}_p([\mathcal{V}_e^{lr}; \mathcal{V}_e^d])])),\end{aligned}\tag{5}$$

where  $\rho(\cdot)$  is the softmax function and outputs are probabilities for 3 severity grades and 3 asymmetry classes. Final predictions are derived via the argmax function. A cross-constraint mechanism enforces clinical consistency between severity and asymmetry predictions. Healthy subjects classified as ‘absence’ in the severity grading task are automatically assigned to the ‘non-asymmetry (NA)’ class in the asymmetry prediction task.

## Shapley value

Shapley values interpret the HGNN by quantifying feature contributions based on cooperative game theory [54], thus assessing the rationale behind the prediction. For a model  $f$  with input  $N = \{x_1, \dots, x_n\}$ , Shapley value  $\varphi_i$  for feature  $x_i$  is defined as

$$\varphi_i = \sum_{S \subseteq N \setminus \{x_i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{x_i\}) - f(S)],\tag{6}$$

where  $S$  is a subset of  $N$  excluding  $x_i$ . Shapley values are computed separately for each of the six classes and normalized to  $[0, 1]$ , where 0 and 1 denote no contribution and maximal contribution to a class, respectively. This enables interpretation of feature importance across expressions, AUs, and asymmetry keys that HGNN learns.

## Statistics and reproducibility

Codes were written in Python 3.8 and PyTorch 1.9.0. Experiments were performed on an NVIDIA RTX 3090 GPU and an Intel Xeon CPU. HGNN was trained by the Adam optimizer using a learning rate of  $1 \times 10^{-4}$  with decay. Training used a class-balanced focal loss with a focusing parameter of  $\gamma = 2$ . We used horizontal flipping augmentation during the training phase. Evaluation was conducted in a 10-fold cross-validation. Individual-wise random splitting (sample size = 98) was employed due to the high phenotypic variability within families. Accuracy, sensitivity, specificity, precision, and the ROC curve were calculated as evaluation metrics. Statistical analyses were conducted using the paired permutation test.

## Standard protocol approvals, registrations, and patient consents

All study procedures were conducted according to the principles of the Declaration of Helsinki and approved by the HKU/Hospital Authority HK West Cluster IRB Committee (UW 19-356, UW 21-009) and HKU Human Research Ethics Committee (EA230139). Written informed consent was obtained from adult participants or legally authorized representatives of the minor participants. The FSHD patients included in this study were screened from the Hong Kong NMD registry cohort.

## Results

### Cohort characteristics

Table 2 summarizes the demographic characteristics of the cohort. Among the 98 participants, 46 were FSHD patients recruited from 31 families. Disease severity and facial weakness patterns varied within families, reflecting the progressive and heterogeneous nature of FSHD. All patients had Type-1 FSHD, with an average of 5.00 *D4Z4* repeat units (SD: 1.62 units).

Table 2: Demographic and clinical characteristics of the study participants (n=98)

|                                  | FSHD patients (n=46) | Healthy controls (n=52) | Total (n=98)    |
|----------------------------------|----------------------|-------------------------|-----------------|
| <b>Demographics</b>              |                      |                         |                 |
| <b>Mean age in years</b>         | 46.2 $\pm$ 17.7      | 30.8 $\pm$ 8.7          | 38.0 $\pm$ 15.7 |
| <b>Female</b>                    | 17 (37.0%)           | 20 (38.5%)              | 37 (37.8%)      |
| <b>Facial Weakness Severity</b>  |                      |                         |                 |
| <b>Absent</b>                    | 5 (10.9%)            | 52 (100%)               | 57 (58.2%)      |
| <b>Moderate</b>                  | 17 (37.0%)           | 0                       | 17 (17.3%)      |
| <b>Severe</b>                    | 24 (52.2%)           | 0                       | 24 (24.5%)      |
| <b>Facial Weakness Asymmetry</b> |                      |                         |                 |
| <b>Non-asymmetry</b>             | 6 (13.0%)            | 52 (100%)               | 58 (59.2%)      |
| <b>Left side</b>                 | 26 (56.5%)           | 0                       | 26 (26.5%)      |
| <b>Right side</b>                | 14 (30.4%)           | 0                       | 14 (14.3%)      |

Fig. 4 visualizes the statistical distributions of eight facial expressions, 17 AUs, and 24 FAs in 98 subjects to provide insights into our cohort characteristics. For facial weakness severity, we set the absence grade as the baseline to analyze the moderate and severe grades. Using the AU distribution (Fig. 4 (a)) as an example, the curve for moderate grade was obtained by subtracting the mean value for each AU in moderate patients from that in absence subjects. As shown in Fig. 4 (c), the expression distribution of patients exhibits a notable distance from the absence baseline. This indicates that patients with and without facial weakness manifest discernible differences when presenting expressions. As shown in Fig. 4(a)–(b), the AU and FA distributions of patients fluctuate around the absence baseline. The distribution curves for the severe grade show greater variation compared with the moderate grade, confirming that severe FSHD patients exhibit greater kinematic variability in their facial behavior.

For facial weakness asymmetry, we set the non-asymmetry subjects as the baseline to analyze feature distributions in left- and right-sided facial weakness. The results revealed a consistent discrepancy between the three FSHD asymmetry types. Almost opposite patterns were observed in the expression (Fig. 4 (f)) and FA (Fig. 4 (e)) distributions. This agrees with the fact that FSHD patients with unilateral facial weakness primarily contract the facial muscles on only one side to perform expressions.

### Assessment performance

For FSHD severity grading (absence vs. moderate vs. severe), HGNN achieved a task-level accuracy of 76.1%, defined as the proportion of subjects assigned the correct severity grade across all three classes. Given the clinical relevance of the individual severity categories, we additionally examined class-specific performance using per-class accuracy, sensitivity, specificity, and precision. The detailed results were summarized in Table 3. It showed a performance gradient as Absence > Severe > Moderate, consistent with the fact that extreme states yield more distinct features and thus a more confident decision boundary. The preserved performance in underrepresented grades still indicated the model’s strong robustness despite class imbalance.

For facial asymmetry prediction, HGNN achieved a task-level accuracy of 74.9%. Table 4 summarizes class-

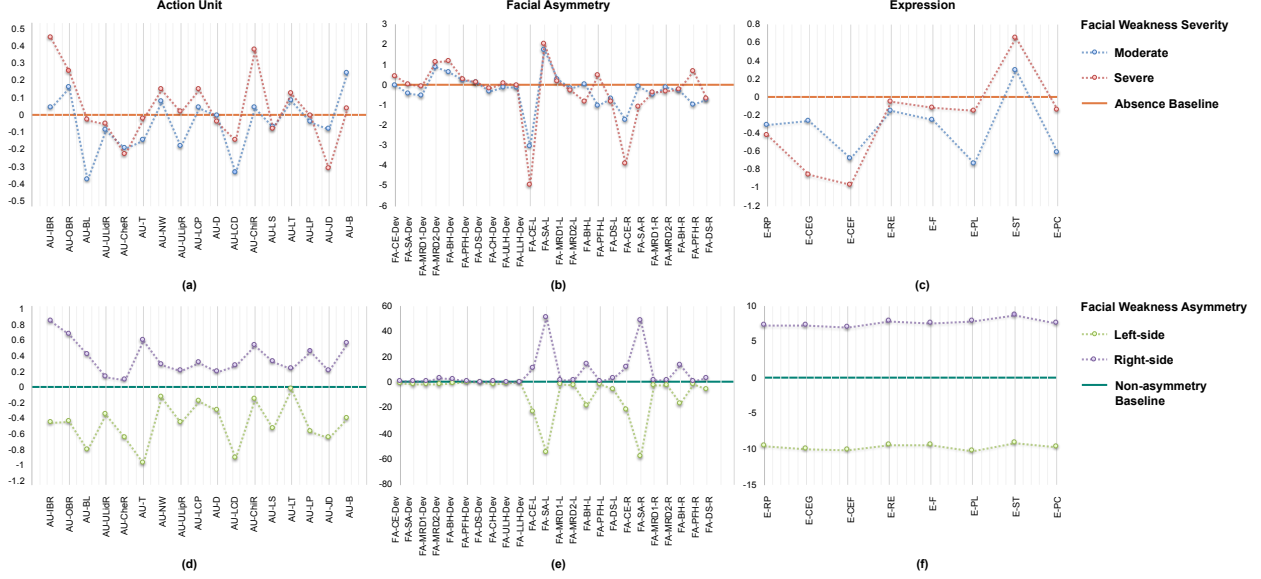


Figure 4: Data distributions (n=98 human subjects). (a-c) 17 action units, 24 asymmetry keys, and 8 expressions in FSHD patients exhibiting moderate and severe facial weakness, relative to an absence-class baseline. (d-f) Distributions of the three data types in FSHD patients with left- versus right-dominant facial asymmetry, relative to the non-asymmetric baseline.

specific accuracy, sensitivity, specificity, and precision. The non-asymmetry class demonstrated superior performance across all four metrics, while the two types of unilateral facial weakness proved to be more challenging.

Fig. 5 shows the receiver operating characteristic (ROC) curves with the area under the curve (AUC). Regarding severity prediction (Fig. 5 (a)), HGNN achieved an average AUC of 0.84 but exhibited higher cross-fold variability on the moderate grade with an AUC standard deviation of 0.22. This elevated variability arises from two factors. First, the moderate grade comprises only 17.3% of the cohort. This class imbalance introduces greater sampling variance during fold stratification. Fewer samples also cause less stable decision boundaries [55, 56]. Second, the moderate grade represents a transitional phenotypic state [21, 22], which creates overlapping features with absence and severe grades. Our model mirrors clinical practice where inter-rater disagreement is most pronounced for intermediate severity [23]. On the other hand, as shown in Fig. 5 (b), the AUC for asymmetry prediction is 0.875, suggesting high discriminative ability for predicting facial weakness asymmetry.

Table 3: Model performance in grading the facial weakness severity of FSHD

| Metrics         | Severity grades of facial weakness in FSHD |                    |                  | Average |
|-----------------|--------------------------------------------|--------------------|------------------|---------|
|                 | Absence<br>(n=57)                          | Moderate<br>(n=17) | Severe<br>(n=24) |         |
| Accuracy* (%)   | 87.8 ± 10.2                                | 80.8 ± 7.9         | 83.5 ± 7.5       | 84.0    |
| Sensitivity (%) | 85.4 ± 11.9                                | 65.3 ± 22.9        | 67.5 ± 29.2      | 72.7    |
| Specificity (%) | 90.0 ± 11.2                                | 84.9 ± 11.2        | 89.8 ± 7.8       | 88.2    |
| Precision (%)   | 91.7 ± 10.4                                | 60.8 ± 22.6        | 76.3 ± 18.3      | 76.3    |

\* Per-class one-vs-rest accuracy.

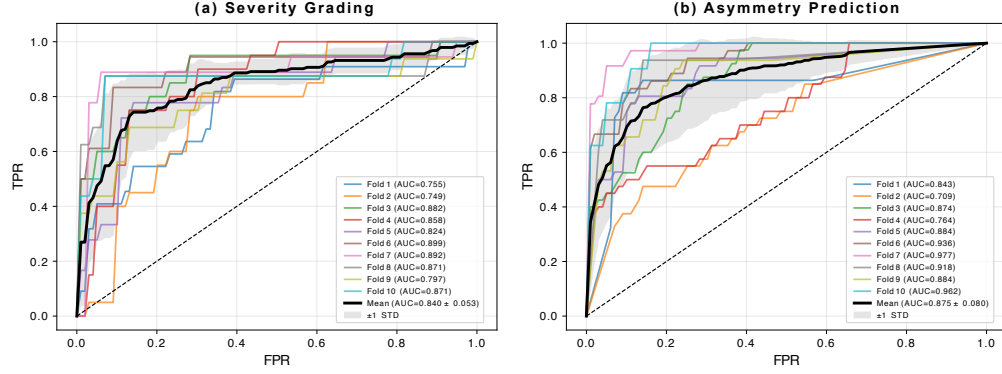


Figure 5: Receiver operating characteristic (ROC) curves for (a) FSHD severity grading and (b) facial asymmetry prediction. Bold black lines denote the average of 10-fold cross-validation, and shaded regions indicate standard deviation.

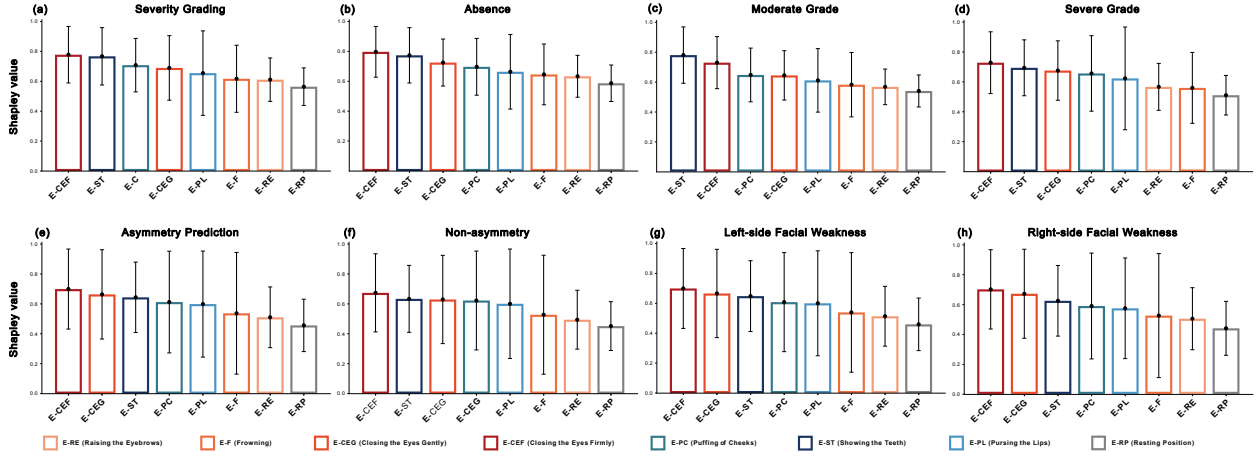


Figure 6: Learned feature importance of the 8 expressions. (a) Aggregated Shapley values for FSHD severity grading. (b-d) Class-specific Shapley values for three severity grades. (e) Aggregated Shapley values for FSHD asymmetry prediction. (f-h) Class-specific Shapley values for non-asymmetry, left-side, and right-side facial weakness. Error bars denote the standard deviation.

Table 4: Model performance in predicting facial weakness asymmetry in FSHD

| Metrics         | Facial sides with higher level of FSHD |                     |                      | Average |
|-----------------|----------------------------------------|---------------------|----------------------|---------|
|                 | Non-asymmetry<br>(n=58)                | Left side<br>(n=26) | Right side<br>(n=14) |         |
| Accuracy*(%)    | 88.1 ± 8.8                             | 80.1 ± 9.7          | 81.6 ± 7.5           | 83.3    |
| Sensitivity (%) | 87.0 ± 12.9                            | 68.0 ± 23.5         | 54.6 ± 22.0          | 69.9    |
| Specificity (%) | 89.4 ± 10.8                            | 83.8 ± 14.7         | 89.6 ± 10.3          | 87.6    |
| Precision (%)   | 91.8 ± 9.2                             | 60.8 ± 17.4         | 68.9 ± 20.6          | 73.8    |

\* Per-class one-vs-rest accuracy.

## Model interpretation

The HGNN was interpreted by the Shapley values to elucidate its decision-making rationale. As shown in Fig. 6 (a), for the severity grading task, HGNN predominantly leveraged orofacial and periocular dynamics, with

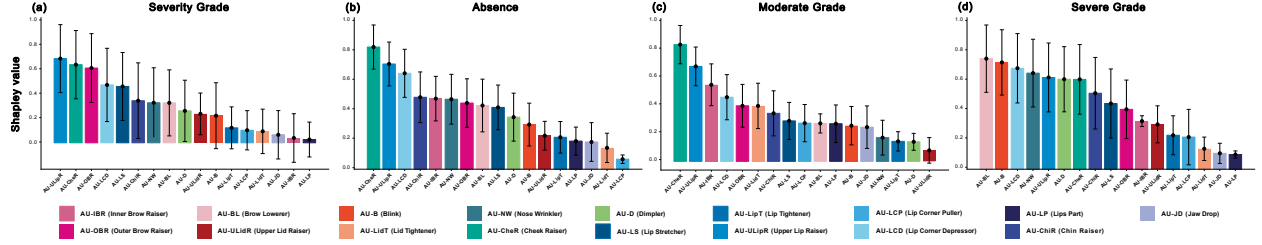


Figure 7: Learned feature importance of the 17 action units. (a) Aggregated Shapley values for FSHD severity grading. (b-d) Class-specific Shapley values for three severity grades. Error bars denote the standard deviation.

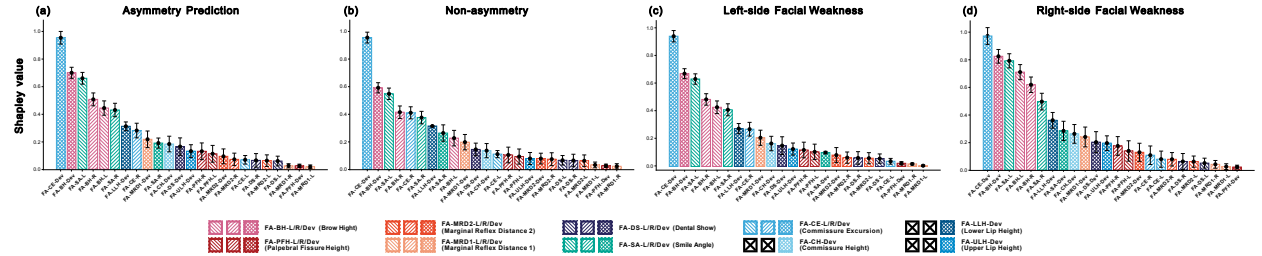


Figure 8: Learned feature importance of the 24 facial asymmetry keys. (a) Aggregated Shapley values for FSHD asymmetry prediction. (b-d) Class-specific Shapley values for non-asymmetry, left-side, and right-side facial weakness. Abbreviations “L” and “R” denote measurements on the left-side and right-side faces, “Dev” denotes the deviation between the unilateral measurements. Error bars denote the standard deviation.

“showing the teeth,” “puffing the cheeks,” and “closing the eyes firmly” emerging as the most discriminative expressions. “showing the teeth” optimally characterized moderate FSHD, whereas forceful eye closure proved most predictive of severe FSHD (Fig. 6(c)–(d)). The higher weighting of forced over gentle eye closure corroborated clinical practices, as forceful closure can better reveal subtle weakness and discriminate patients with Bell’s phenomenon [46]. These preferential expressions were associated with the actions of the lip, cheek, and brow, corresponding to the action units with the highest contribution in Fig. 7 (a). The “cheek raiser” and “brow lowerer” were the most indicative action units for moderate and severe FSHD (Fig. 7 (c)–(d)).

As shown in Fig. 6 (e), for asymmetry prediction, both “firmly and gently closing the eyes”, as well as “showing the teeth”, remained highly informative. Comparing Fig. 6 (g) and (h), left- and right-side facial weakness exhibited a consistent prioritization of expressions. This emphasis on peri-oral and peri-ocular expressions closely mirrored the pathognomonic phenotype of FSHD, where asymmetries of the oral commissures and eyebrows were the hallmark clinical signatures of FSHD [23].

As shown in Fig. 8, HGNN consistently prioritized “commissure excursion,” “brow height,” and “smile angle” as the most salient asymmetry keys for the facial asymmetry prediction task. Notably, “commissure excursion” stood out as the most diagnostically valuable indicator for all asymmetry conditions (Fig. 8(b)–(d)). These interpretation results confirmed that HGNN learned physiologically grounded FSHD facial assessment.

## Method comparison

We conducted a comparative study evaluating HGNN against three machine learning methods (XGBoost [57], Random Forest [58], SVM [59]) and four non-graph neural networks (FT-Transformer [60], 1D CNN [61], 2D CNN [62], FFNN [63]) under identical experimental conditions. Table 5 demonstrates that HGNN achieved the highest averaged per-class accuracy in both severity grading and asymmetry prediction tasks

and the most balanced performance across four metrics.

Table 5: Comparison between HGNN and the other seven methods on two tasks

| Comparison Method           |                | Severity Grading |           |           |           | Asymmetry Prediction |           |           |           |
|-----------------------------|----------------|------------------|-----------|-----------|-----------|----------------------|-----------|-----------|-----------|
|                             |                | Acc.*            | Sen.      | Spe.      | Pre.      | Acc.*                | Sen.      | Spe.      | Pre.      |
| Machine Learning            | XGBoost        | 81.4±4.5         | 60.7±32.4 | 84.4±8.1  | 62.9±23.3 | 79.9±5.1             | 55.6±28.0 | 84.0±4.8  | 58.9±18.4 |
|                             | Random Forest  | 83.0±1.4         | 61.6±35.7 | 83.8±15.1 | 77.9±0.3  | 79.2±0.9             | 55.7±31.2 | 80.3±16.9 | 65.0±7.7  |
|                             | SVM            | 78.9±4.0         | 59.4±27.8 | 83.0±4.0  | 65.1±15.2 | 78.5±3.7             | 59.4±21.2 | 83.0±3.5  | 71.8±9.7  |
| Non-graph<br>Neural Network | FT-Transformer | 78.2±3.7         | 59.5±22.2 | 80.0±15.6 | 70.1±6.4  | 74.3±2.4             | 50.7±25.7 | 76.3±18.0 | 59.2±8.2  |
|                             | CNN-1D         | 79.7±3.5         | 60.9±25.6 | 83.6±4.1  | 63.3±16.5 | 77.7±3.9             | 58.5±19.3 | 82.9±1.6  | 60.1±17.3 |
|                             | CNN-2D         | 80.1±3.2         | 61.2±25.1 | 83.9±5.0  | 65.6±14.7 | 80.5±2.6             | 63.3±20.5 | 84.1±7.5  | 69.7±12.2 |
|                             | FFNN           | 82.3±4.5         | 65.5±25.9 | 85.9±2.9  | 69.0±16.0 | 81.5±3.1             | 65.6±17.4 | 85.6±2.6  | 69.8±12.9 |
| Ours                        | HGNN           | 84.0±2.9         | 72.7±9.0  | 88.2±2.4  | 76.3±12.6 | 83.3±3.5             | 69.9±13.3 | 87.6±2.7  | 73.8±13.1 |

\* Averaged per-class accuracy.

Machine learning methods exhibit divergent behavior. XGBoost achieved high specificity but the lowest precision in both tasks, indicating less reliable assignment to specific severity and asymmetry categories despite effective exclusion of non-target cases. Random Forest achieved competitive accuracy but yielded low sensitivity paired with large variance. SVM also demonstrated relatively high specificity, but its performance across other metrics was lower. These machine learning methods thus present a potential risk of missed diagnoses and performance instability. Non-graph neural networks exhibited comparable performance but consistently fell short of HGNN across all metrics. This suggests that data-driven models without clinical priors imposed by graph constraints can fail to capture pathophysiological relationships among facial features for FSHD phenotyping.

## Ablation study

To validate the necessity of each graph component, we conducted an ablation study wherein the  $\mathcal{G}_e$ ,  $\mathcal{G}_{au}$ , and  $\mathcal{G}_{fa}$  were sequentially removed. As Table 6 shows,  $\mathcal{G}_e$  alone yielded a baseline accuracy of about 78% on both tasks. Integrating  $\mathcal{G}_{au}$  elevated severity grading performance, attributable to its localized modeling of anatomical muscle synergies. The inclusion of  $\mathcal{G}_{fa}$  further optimized lateralized weakness quantification. The complete HGNN achieved superior performance across all metrics, underscoring the non-redundant and complementary nature of the proposed graphs.

Table 6: Performance metrics for graph ablation on two tasks

| Graph           |                    |                    | Severity Grading |           |          |           | Asymmetry Prediction |           |          |           |
|-----------------|--------------------|--------------------|------------------|-----------|----------|-----------|----------------------|-----------|----------|-----------|
| $\mathcal{G}_e$ | $\mathcal{G}_{au}$ | $\mathcal{G}_{fa}$ | Acc.*            | Sen.      | Spe.     | Pre.      | Acc.*                | Sen.      | Spe.     | Pre.      |
| ✓               |                    |                    | 78.0±2.4         | 63.1±13.9 | 83.2±1.5 | 67.2±16.7 | 78.1±2.1             | 61.9±15.4 | 82.7±4.9 | 67.5±12.9 |
| ✓               | ✓                  |                    | 81.9±2.7         | 68.5±15.2 | 86.5±2.8 | 69.8±17.1 | 78.5±5.6             | 60.0±22.9 | 84.1±5.7 | 64.3±17.9 |
| ✓               |                    | ✓                  | 78.7±4.1         | 61.6±15.6 | 84.3±1.5 | 64.2±16.9 | 79.9±4.1             | 62.9±17.8 | 84.8±1.0 | 68.8±13.6 |
|                 | ✓                  | ✓                  | 79.1±1.9         | 66.9±11.6 | 85.3±4.6 | 69.1±16.7 | 75.4±5.7             | 58.0±14.3 | 82.3±8.9 | 64.6±19.9 |
| ✓               | ✓                  | ✓                  | 84.0±2.9         | 72.7±9.0  | 88.2±2.4 | 76.3±12.6 | 83.3±3.5             | 69.9±13.3 | 87.6±2.7 | 73.8±13.1 |

\* Averaged per-class accuracy.

Ablation studies also evaluated the contribution of the GGA and cross constraint. As shown in Table 7, removing GGA led to degradation across both tasks, confirming its critical role in integrating global contextual cues to refine local graph representations. Ablating the cross constraint also yielded performance drops due to the lack of task-consistent constraints. These results demonstrated that these modules jointly ensured robust and clinically faithful facial assessment.

Table 7: Performance metrics for module ablation on two tasks

| Ablation            | Severity Grading |                 |                |                 | Asymmetry Prediction |                 |                |                 |
|---------------------|------------------|-----------------|----------------|-----------------|----------------------|-----------------|----------------|-----------------|
|                     | Acc.*            | Sen.            | Spe.           | Pre.            | Acc.*                | Sen.            | Spe.           | Pre.            |
| w/o GGA             | 80.1 $\pm$ 2.0   | 67.5 $\pm$ 7.7  | 85.5 $\pm$ 4.4 | 68.5 $\pm$ 17.7 | 80.6 $\pm$ 3.6       | 65.7 $\pm$ 13.6 | 85.6 $\pm$ 1.8 | 67.3 $\pm$ 15.1 |
| w/o Cross Constrain | 82.8 $\pm$ 3.0   | 67.7 $\pm$ 16.9 | 86.5 $\pm$ 3.2 | 71.8 $\pm$ 11.2 | 82.3 $\pm$ 3.1       | 67.0 $\pm$ 19.5 | 86.1 $\pm$ 4.4 | 72.4 $\pm$ 11.6 |

\* Averaged per-class accuracy.

## Discussion

This study proposes a graph-based model for FSHD facial assessment and delivers a dedicated dataset for this rare disease. In addition to severity grading, this study establishes a multi-task joint learning framework that incorporates the asymmetry assessment. Our findings demonstrate the feasibility and interpretability of the model for predicting both severity and asymmetry of facial weakness in FSHD.

Currently, there are still gaps in tools for diagnosing facial weakness in FSHD [23, 27]. For instance, the clinician-reported 4-point facial weakness score shows low interrater agreement [23], and the patient-reported Facial Function Scale may require additional cross-cultural and language validation [27]. Our method offers an objective, patient-friendly, and interpretable approach to FSHD-related facial weakness assessment. Specifically, HGNN automates the expert-based facial weakness assessment in alignment with the standardized CCEF scale, thereby streamlining the current diagnostic process. Because the model requires only static facial photographs that can be captured with consumer-level cameras, this assessment becomes feasible outside specialist centers—including at-home self-measurement—without imposing additional burden on the healthcare system or compromising patient privacy. Such accessibility may further support population-level screening for earlier intervention in clinical settings lacking neuromuscular expertise.

While end-to-end approaches directly processing images or videos represent an alternative for facial analysis [32, 64], our study prioritized patient privacy given the sensitive nature of facial data. Patients with FSHD often experience heightened psychosocial stress related to facial appearance [19], making the dataset of raw photographic data ethically unavailable. By utilizing de-identified facial landmarks to encode expression, action, and asymmetry features, we demonstrate robust diagnostic performance without compromising patient privacy.

The comparison study demonstrates that graph-based feature processing yields more discriminative representations than convolutional or fully connected alternatives. Embedding clinical priors through graph topology guides feature aggregation along physiologically meaningful pathways, thus enhancing generalization in the current sample size of FSHD cohorts. The limited performance of attention-based architectures underscores the challenge of deploying high-capacity models to data-scarce tasks without graph regularization.

Given that current FSHD scoring practices involve facial assessment using standardized facial expressions, the proposed model can be seamlessly integrated into this clinical workflow, providing objective, automated severity grading and asymmetry prediction based on routine photographs. Prior to clinical deployment, rigorous clinical trials are required for longitudinal validation. Future trials will correlate the model-derived predictions with existing diagnostic markers, including muscle biopsy, functional muscle evaluations, and quantitative imaging, to establish HGNN as a reliable screening tool from disease onset. While our model quantifies functional severity and asymmetry of facial weakness, future studies will correlate these objective metrics with patient-reported outcome measures (PROMs, e.g., Facial Disability Index [65]) and biomechanical measures to strengthen its clinical interpretability by linking functional impact to biomechanical impairment and quality of life [66].

Recent work by Loonen *et al.* [32] provides an important complementary approach by generating continuous measurements of facial weakness, which is advantageous for detecting longitudinal changes. In contrast, our model performs facial severity and asymmetry classification, maintaining correspondence with current

clinical practice. Because a widely validated continuous scale for FSHD facial weakness is not yet available, we formulated our task as classification using consensus-based ordinal labels, consistent with established standards [21, 22]. Future studies will extend this framework to continuous measurement once a reliable continuous reference is ready. Our use of 2D photographs simplifies data acquisition. However, videos of facial expressions provide richer information about muscle dynamics and can better capture subtle abnormalities. Future work will therefore focus on extending our method using video-derived facial dynamics.

This proof-of-principle study has several limitations. The cross-sectional data collection protocol did not include scheduled serial visits or assessments before and after treatment. Therefore, responsiveness to change and sensitivity to disease progression or treatment remain to be evaluated. Classification based on static photographs may have limitations in capturing subtle changes within a severity category and needs to be tested with a larger number of patients for its potential use as a longitudinal outcome measure. Future prospective studies will prioritize standardized follow-up and treatment-response assessments.

The generalizability of the current findings is constrained by characteristics of the study cohort. There is a demographic mismatch between the age of healthy controls and patients. Age-related changes in soft-tissue volume, skin elasticity, and wrinkle patterns may affect action and asymmetry quantification. The younger healthy cohort presented fewer confounding variations, potentially making healthy classification easier. The modest sample size also partly accounts for lower accuracy in the subgroup with fewer subjects [36]. As this study was conducted in a single Chinese cohort, the model’s generalizability to other racial/ethnic populations remains to be established [47]. Multi-center studies with larger and balanced cohorts are needed for comprehensive model evaluation. This study exclusively included Type-1 FSHD patients due to the low prevalence of Type-2 (< 5% in our locality). Future work can prioritize recruiting Type-2 subjects for cross-subtype validation. In this study, model interpretability refers to input feature-level contribution quantified by Shapley value. While the Shapley value provides clinically auditable insight into the rationale behind model predictions, it does not yet translate into actionable thresholds or continuous monitoring scales.

In summary, this study demonstrated the feasibility of automatic and interpretable facial weakness assessment in FSHD using facial photographs in 8 standardized expressions. Our graph-based model achieved high performance in facial weakness severity and asymmetry prediction, with the advantage of simpler data acquisition compared to functional ultrasound and MRI imaging. Importantly, the model learns appropriate importance for different expressions, actions, and asymmetry features, consistent with clinical rationale.

## Data availability

The data collected in this study are not publicly available due to patient privacy regulations and ethical restrictions imposed by the Hospital Authority HK West Cluster IRB Committee and HKU Human Research Ethics Committee to protect participant confidentiality. De-identified sample is available from the corresponding author upon reasonable request. Access will be granted to researchers who provide a methodologically sound proposal and sign a Data Use Agreement that prohibits the re-identification of participants. The corresponding author will respond to access requests within four weeks. The source data for Figures 4-8 are provided in Supplementary Data 1.

## Code availability

The code used in this work is publicly available at <https://github.com/JustinJZhang/HGNN-Hierarchical-Graph-Neural-Network>.

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## Author contributions

S.C. proposed and conceptualized the project with W.L.; J.Z. and A.W. designed the methodology and wrote the algorithm code; M.Y., J.Z., A.W., and C.Y. designed the experimental setup and carried out the data acquisition with the patient cohort from S.C.; J.Z. and J.A.E.P. carried out the data analysis; W.L. and S.C. supervised the project. All authors contributed to the writing of the manuscript.

## Competing interests

The authors declare no competing interests.

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## Figure captions

Fig. 1: Overview of the proposed method for facial weakness assessment in FSHD. The pipeline involves photo collection, landmark detection, feature encoding, a hierarchical graph neural network (HGNN) for multi-task prediction, and Shapley value-based model interpretation.

Fig. 2: The definition of three graphs. (a) Expression graph, where “E-xxx” nodes stand for 8 expressions. (b) Action graph, where “AU-xxx” nodes represent 17 action units. (c) Asymmetry graph, where “FA-xxx-L”, “FA-xxx-R”, and “FA-xxx-Dev” nodes represent 7 left-side, 7 right-side, and 10 deviation asymmetry keys.

Fig. 3: Hierarchical-graph neural network. The inputs are action units and asymmetry keys of 8 expressions, represented as the action graph  $\mathcal{I}_{au}$ , asymmetry graph  $\mathcal{I}_{fa}$ , and expression graph  $\mathcal{I}_e$  for the action path ( $\mathcal{F}_{au}$ ), asymmetry path ( $\mathcal{F}_{fa}$ ), and expression path (including action  $\mathcal{F}_e^{au}$ , deviation  $\mathcal{F}_e^d$ , and left- and right-side  $\mathcal{F}_e^{lr}$  branches).  $|\phi_e| = 8$ ,  $|\phi_{au}| = 17$ , and  $|\phi_{fa}| = 24$  denote the numbers of expression, action, and asymmetry nodes, respectively.  $N_i = 32, 64, 128$  and  $N_p = 128$  denote the numbers of output features of the  $i$ -th graph convolutional layer and the projector, respectively.

- Fig. 4: Data distributions ( $n = 98$  human subjects). (a–c) Distributions of 17 action units, 24 asymmetry keys, and 8 expressions in FSHD patients exhibiting moderate and severe facial weakness, relative to an absence-class baseline. (d–f) Distributions of the three data types in FSHD patients with left-versus right-dominant facial asymmetry, relative to the non-asymmetric baseline.
- Fig. 5: Receiver operating characteristic (ROC) curves for (a) FSHD severity grading and (b) facial asymmetry prediction. Bold black lines denote the averages obtained from 10-fold cross-validation, and the shaded regions indicate the standard deviations.
- Fig. 6: Learned feature importance of the 8 expressions. (a) Aggregated Shapley values for FSHD severity grading. (b–d) Class-specific Shapley values for the three severity grades. (e) Aggregated Shapley values for FSHD asymmetry prediction. (f–h) Class-specific Shapley values for non-asymmetry and left- and right-side facial weakness. Error bars denote the standard deviations.
- Fig. 7: Learned feature importance of the 17 action units. (a) Aggregated Shapley values for FSHD severity grading. (b–d) Class-specific Shapley values for the three severity grades. Error bars denote the standard deviations.
- Fig. 8: Learned feature importance of the 24 facial asymmetry keys. (a) Aggregated Shapley values for FSHD asymmetry prediction. (b–d) Class-specific Shapley values for non-asymmetry, left-side, and right-side facial weakness. Abbreviations “L” and “R” denote measurements on the left-side and right-side faces, “Dev” denotes the deviation between the unilateral measurements. Error bars denote the standard deviation.